

Zack LaPrade

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EDUCATION

University of Georgia	Athens, GA
<i>Bachelor of Science, Computer Engineering</i>	May 2027
Scholarships: Zell Miller Scholarship	Cumulative GPA: 3.54/4.00

WORK AND RESEARCH EXPERIENCE

Technology Service Corporation - Defense Contractor	Huntsville, AL
<i>Electrical Engineering Intern</i>	May - August 2026

- Researched and tested an Over-Current Detection device, detecting over-current in devices within 300ns
- Created schematics and PCB Layout for Over-Current Detection and several other schematics
- Debugged and validated units through measures such as X-Rays, probing voltages/resistances, and replacing components
- Gained experience using radio frequency and a Network Analyzer to scan the Quiet Zone of a compact range

UGA Robotics Research	Athens, GA
<i>Undergraduate Researcher</i>	February 2026 - Present

- Developed a glove that allows for control of another robotic hand across a 40-foot room with a response of less than 80 ms
- Implemented the required sensors, including 5 FSRs and an IMU, to develop the project to allow gripping and rotation

Peer Learning Assistant	Athens, GA
<i>UGA Physics PLA</i>	August - December 2025

- Assisted the professor in answering questions/misconceptions from students in Physics 1 in lecture and/or lab
- Attended weekly peer education classes to practice effective communication methods when teaching and instructing

PROJECT EXPERIENCE

PepsiCo Capstone Project	August 2026 - Present
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- Analyzing fault logs and alarm codes across 25-30 LGVs to identify high-impact failures for a PepsiCo warehouse
- Mapping LGV fault locations across a PepsiCo warehouse to correlate failures with routes and workload

Raspberry Pi Smart Room System	May 2025 - Present
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- Hosted a local server on a Raspberry Pi, logged sleep, hydration, exercise, and temperature to self-identify my migraine triggers and prevent them
- Utilized 4 API's, 20 threads, 21 methods, 3 sensors and 2 sockets to add room data, lightbulbs, and health data
- Achieved a hands-free and automatic experience through 4 different mediums: a Bluetooth button, voice commands, an iPhone app, and a wall display
- Applied offline speech recognition, sensor readings and API calls are controlled without internet or cloud service dependency

Multitool Bench	January - May 2026
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- Worked with 4 classmates as hardware lead to replicate standard tools without ICs and achieved all 14 checkpoints
- Derived a stable +-12V supply through a class B push-pull amplifier with feedback to ensure no voltage sag
- Designed a Voltmeter and Ohmmeter using a 6-bit ADC and a Wheatstone bridge to negate noise with 2% error
- Created a frequency counter for any waveform utilizing a transistor with accuracy to 0.1 Hz

Smart Water Bottle	January - August 2026
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- Integrated multiple sensors to measure water temperature, capacitance of the water, and water flow from the bottle for automatic logging and coaching on hydration
- Deployed HTTPS, TCP, and I2C to connect for sensor fusion with a 99.1% accuracy and 20ms response time
- Utilized AutoCAD to design and 3D print a chassis to house the ESP32, sensors, and battery

MEMBERSHIPS AND ASSOCIATIONS

Theta Tau, Professional Engineering Fraternity	Athens, GA
<i>Rush / Recruitment Chair</i>	May 2025 - May 2026

- Collaborated with 2 co-chairs to plan and advertise all rush events and interviews for 250+ rushees
- Gained experience in public speaking, communication, and improved time management to effectively meet deadlines

IEEE, Institute of Electrical and Electronic Engineers	
<i>Member</i>	August 2025 - Present

- Engaged in technical workshops and projects to enhance engineering skills, professionalism, and communication

TECHNICAL SKILLS AND LANGUAGES

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- **Technical:** Matlab, Altium, Java, C, C++, Python, HTML, JavaScript, Parallel Programming, Circuit Analysis, Multisim
 - **Languages:** English (Fluent), Spanish (Advanced), French (Advanced)

AWARDS AND ACCOLADES

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- International Baccalaureate Diploma, International Skills Diploma Seal, Georgia Seal of Biliteracy